

Andrew McNutt

Assistant Professor of HCI & Visualization

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mcnutt.in

Education

University of Chicago

Ph.D., Computer Science (*Advisor Ravi Chugh*)

Masters of Science, Computer Science (*Advisor Gordon Kindlmann*)

Chicago, IL

2017 to 2023

2017 to 2019

Reed College

Bachelor of Arts, Physics (*Advisor Nelia Mann*)

Portland, OR

2010 to 2014

Professional & Research Experience

- 2024 to **University of Utah**, *Assistant Professor*
2023 to 2024 **University of Washington**, *Post Doctoral Scholar*
 Supervised by Jeffrey Heer and Leilani Battle
- 2022 **Microsoft Research**, *Intern*
 Supervised by Chenglong Wang, Rob DeLine, and Steven M. Drucker
- 2019 **Tableau Research**, *Intern*
 Supervised by Michael Correll
- 2015 to 2017 **Uber**, *Data Visualization Engineer II*
2014 to 2015 **Collaborative Drug Discovery**, *Scientific Visualization Developer*

Funding

- 2024 **NSF CISE Core III**: Medium: Towards Provably Effective Visualization Design.
 PI Battle, Co-PIs McNutt, Heer (Utah \$400k / Mine \$400k / Full \$1.2m)
- 2023 **Postdoc Research Award** *University of Washington* (~\$7,000)
- 2022 **Doctoral Colloquium Fellowship** . *IEEE VIS* (~\$4,000).
- 2021 **Graduate Council Research Fund**. *UChicago Grad* (\$400).
- 2020 **Graduate Council Travel Fund**. *UChicago Grad* (\$400).

Awards

- 2025 **Best Paper**. *IEEE VIS* for *ReVISit 2* (VIS25)
 Best Paper Honorable Mention. *ACM SIGCHI* for *Slowness, Politics, and Joy* (CHI25)
- 2023 **Siebel Scholars Class of 2023 (\$35,000)**. *Siebel*.
- 2022 **Long List in the Business Intelligence Category**. *Information is Beautiful Awards*.
- 2021 **Best Paper Honorable Mention**. *EuroVIS* for *What are Table Cartograms Good for Anyway?*
 (*EuroVis21*)
- 2020 **Best Poster Honorable Mention**. *IEEE VIS* for *A Minimally Constrained Optimization Algorithm for Table Cartograms* (VIS20)
 Best Paper Honorable Mention. *ACM SIGCHI* for *Surfacing Visualization Mirages* (CHI20)
- 2019 **Best Poster in Visualization (~\$200)**. *MindBytes Research Symposium*.
 Long List for Visual Analytics and Unusual Categories. *Information is Beautiful Awards*.
 Teaching Assistant Prize. *University of Chicago, Department of Computer Science*.
 2nd Place for Best in Show. *UChicago Art and Science Expo*.
- 2019- **11 Special Recognitions for Outstanding Reviews**. *CHI, VIS, UIST*
2014 **Commendation of Academic Excellence**. *Reed College*.

Underline indicates University of Utah student

- 2026 **Linting Style and Substance in READMEs**
Hima Mynampaty, Nathania Josephine, Katherine E. Isaacs, **Andrew McNutt**. ACM Conference on Human Factors in Computing Systems (SIGCHI). 2026
- Same Data, Different Audiences: Using Personas to Scope a Supercomputing Job Queue Visualization**
Connor Scully-Allison, Kevin Menear, Kristin Potter, **Andrew McNutt**, Katherine E. Isaacs, Dmitry Duplyakin. IEEE Transactions on Visualization and Computer Graphics. 2026
- Towards Scalable Visual Data Wrangling via Direct Manipulation**
El Kindi Rezig, Mir Mahathir Mohammad, Nicolas Baret, Ricardo Mayerhofer, **Andrew McNutt**, Paul Rosen. Conference on Innovative Data Systems Research (CIDR). 2026
- 2025 **ReVISit 2: A Full Experiment Life Cycle User Study Framework**
Zach Cutler, Jack Wilburn, Hilson Shrestha, Yiren Ding, Brian C Bollen, Khandaker Abrar Nadib, Tingying He, **Andrew McNutt**, Lane Harrison, Alexander Lex. IEEE Transaction on Visualizations and Computer Graphics (proceedings of VIS). Nov. 2025 ★ Best Paper ★ (Top 5/666)
- Accessible Text Descriptions for UpSet Plots**
Andrew McNutt, Maggie K. McCracken, Ishrat Jahan Eliza, Daniel Hajas, Jake Wagoner, Nate Lanza, Jack Wilburn, Sarah Creem-Regehr, Alexander Lex. Computer Graphics Forum (proceedings of EuroVIS). June 2025
- Slowness, Politics, and Joy: Values That Guide Technology Choices in Creative Coding Classrooms** **Andrew McNutt**, Sam Cohen, Ravi Chugh. ACM Conference on Human Factors in Computing Systems (SIGCHI). May 2025 ★ Honorable Mention for Best Paper ★ (top 5%)
- 2024 **Mixing Linters with GUIs: A Color Palette Design Probe**
Andrew McNutt, Maureen Stone, Jeffrey Heer. IEEE Transaction on Visualizations and Computer Graphics (proceedings of VIS). Oct. 2024
- What Can Interactive Visualization do for Participatory Budgeting in Chicago?**
Alex Kale, Danni Liu, Maria Gabriela Ayala, Harper Schwab, **Andrew McNutt**. IEEE Transaction on Visualizations and Computer Graphics (proceedings of VIS). Oct. 2024
- Considering Visualization Example Galleries**
Junran Yang, **Andrew McNutt**, Leilani Battle. IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC). Sept. 2024
- How Do Data Analysts Respond to AI Assistance? A Wizard-of-Oz Study**
Ken Gu, Madeleine Grunde-McLaughlin, **Andrew McNutt**, Jeffrey Heer, Tim Althoff. ACM Conference on Human Factors in Computing Systems (SIGCHI). May 2024
- 2023 **Projectional Editors for JSON DSLs**
Andrew McNutt, Ravi Chugh. IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC). Oct. 2023
- Metrics-Based Evaluation and Comparison of Visualization Notations**
Nicolas Kruchten, **Andrew McNutt**, Michael McGuffin. IEEE Transaction on Visualizations and Computer Graphics (proceedings of VIS). Oct. 2023

On the Design of AI-powered Code Assistants for Notebooks

Andrew McNutt, Chenglong Wang, Rob DeLine, Steven M. Drucker. ACM Conference on Human Factors in Computing Systems (SIGCHI). April 2023

A Study of Editor Features in a Creative Coding Classroom

Andrew McNutt, Anton Outkine, Ravi Chugh. ACM Conference on Human Factors in Computing Systems (SIGCHI). April 2023

Doom or Deliciousness: Challenges and Opportunities for Visualization in the Age of Generative Models.

Victor Schetinger, Sara Di Bartolomeo, Mennatallah El-Assady, **Andrew McNutt**, Matthias Miller, João Paulo Apolinário Passos, Jane L. Adams. Computer Graphics Forum (proceedings of EuroVIS) 2023.

2022 No Grammar to Rule Them All: A Survey of JSON-style DSLs for Visualization

Andrew McNutt. IEEE Transaction on Visualizations and Computer Graphics (proceedings of VIS). Oct. 2022.

Goethe and Candolle: National Forms of Scientific Writing?

Agatha Kim, **Andrew McNutt**. Theory in Biosciences. August 2022.

Explaining Why: How Instructions and User Interfaces Impact Annotator Rationales When Labeling Text Data.

Jamar L. Sullivan Jr., Will Brackenbury, Kevin Bryson, Kwam Byll, **Andrew McNutt**, Yuxin Chen, Michael L. Littman, Chenhao Tan, Blase Ur. Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL). July 2022.

2021 KondoCloud: Improving Information Management in Cloud Storage via Recommendations Based on File Similarity.

Will Brackenbury, **Andrew McNutt**, Kyle Chard, Aaron Elmore, Blase Ur. ACM Symposium on User Interface Software and Technology (UIST). Oct. 2021.

What are Table Cartograms Good for Anyway? An Algebraic Analysis

Andrew McNutt. Computer Graphics Forum (proceedings of EuroVis). June 2021. ★ Honorable Mention for Best Paper ★ (juried — top 2)

Integrated Visualization Editing via Parameterized Declarative Templates

Andrew McNutt, Ravi Chugh. ACM Conference on Human Factors in Computing Systems (SIGCHI). May 2021.

2020 Surfacing Visualization Mirages

Andrew McNutt, Gordon Kindlmann, Michael Correll. ACM Conference on Human Factors in Computing Systems (SIGCHI). April 2020. ★ Honorable Mention for Best Paper ★ (top 5%)

2015 The Schrodinger-Newton System with Self-field Coupling

Joel Franklin, Youdan Guo, **Andrew McNutt**, Allison Morgan. Journal of Classical and Quantum Gravity. 2015.

Open source Bayesian models. 1. Application to ADME/Tox and Drug Discovery Datasets

Alex M. Clark, Krishna Dole, Anna Coulon-Spektor, **Andrew McNutt**, George Grass, Joel S. Freundlich, Robert C. Reynolds, Sean Ekins. Journal of Chemical Information and Modeling. 2015.

Peer Reviewed Short Papers, Extended Abstracts, and Demos

- 2025 **AnnoGram: An Annotative Grammar of Graphics Extension**
Md Dilshadur Rahman, Md Rahat-uz- Zaman, **Andrew McNutt**, Paul Rosen. Proceedings of IEEE VIS, Short Papers. Nov. 2025.
- Buckaroo: A Direct Manipulation Visual Data Wrangler**
Annabelle Warner, **Andrew McNutt**, Paul Rosen, El Kindi Rezig. VLDB Demo. Sept. 2025. Demo
- Keyframer: A Design Probe for Exploring LLM Assistance in 2D Animation Design**
Tiffany Tseng, Ruijia Cheng, **Andrew McNutt**, Jeffrey Nichols. IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC) (Short). Sept. 2025.
- Linting is People! Exploring the Potential of Human Computation as a Sociotechnical Linter of Data Visualizations**
Anamaria Crisan, **Andrew McNutt**. alt.CHI. April 2025. Extended Abstract.
- 2021 **On The Potential of Zines as a Medium for Visualization**
Andrew McNutt. Proceedings of IEEE VIS, Short Papers. Oct. 2021.
- 2020 **Divining Insights: Visual Analytics Through Cartomancy**
Andrew McNutt, Anamaria Crisan, Michael Correll. alt.CHI. April 2020. Extended Abstract.

Peer Reviewed Workshop Papers and Posters

- 2026 **Literate Execution**
Joseph Bond, Jacob Pake, Cristina David, **Andrew McNutt**, Trevor Muwonge, Dominic Orchard, Roly Perera. PLATEAU Workshop. March 2026. Workshop Paper
- 2025 **Teaching Critical Visualization: A Field Report.**
Andrew McNutt, Shiyi He, Sujit Kumar Kamaraj, Purbid Bambroo, Nastaran Jadidi, John Bovard, Chang Han. EduVis. Nov. 2025. Workshop Paper
- Charts and Measures**
Andrew McNutt. alt.vis. Nov. 2025. Workshop Paper
- The Feng Shui of Visualization: Design the Path to SUCCESS and GOOD FORTUNE**
Chang Han, **Andrew McNutt**. alt.vis. Nov. 2025. Workshop Paper
- Language-Based Dependency-Tracking for Explorables**
Roly Perera, Joseph Bond, Cristina David, **Andrew McNutt**, Alfonso Piscitelli. LIVE Workshop. Sept. 2025. Workshop Presentation
- Authoring Tools for Transparent Climate Reporting**
Roly Perera, Joseph Bond, Cristina David, **Andrew McNutt**, Alfonso Piscitelli. ACM SIGPLAN International Workshop on Programming for the Planet. Oct. 2025. Abstract
- Critical Data Visualization—Part I/II**
Georgia Panagiotidou, **Andrew McNutt**, Derya Akbaba, Nicole Hengesbach, Miriah Meyer. Computer Graphics & Applications. Summer. 2025. Special Issue(s) Introduction
- ReVISitPy: Python Bindings for the reVISit Study Framework**

Hilson Shrestha, Jack Wilburn, Brian Bollen, **Andrew McNutt**, Alexander Lex, Lane Harrison. EuroVIS Posters. June 2025. Poster.

2023 [Only YOU Can Make IEEE VIS Environmentally Sustainable](#)
Elsie Lee-Robbins, **Andrew McNutt**. alt.vis. Oct. 2023. Workshop Paper.

2021 [Visualization for Villainy](#)
Andrew McNutt, Lilian Huang, Kathryn Koenig. alt.vis. Oct. 2021. Workshop Paper.

2020 [Supporting Expert Close Analysis of Historical Scientific Writings: A Case Study for Near-by Reading](#) **Andrew McNutt**, Agatha Kim, Sergio Elahi, Kazutaka Takahashi. IEEE VIS Visualization for the Digital Humanities Workshop. Oct. 2020. Workshop Paper.

[A Minimally Constrained Optimization Algorithm for Table Cartograms](#)

Andrew McNutt, Gordon Kindlmann. InfoVis Posters. Oct. 2020. Poster. ★ Honorable Mention for Best Poster ★ (juried — top 3)

2019 [Textual Analysis & Comparison National Forms of Scientific Texts: Goethe + de Candolle](#)
Agatha Kim, **Andrew McNutt**, S. Sergio Elahi, Kazutaka Takahashi, Robert J Richards. MindBytes Research Symposium. November 2019. Poster. ★ Best Poster in Visualization ★

[Improving the Scalability of Interactive Visualization Systems for Exploring Threaded Conversations](#). **Andrew McNutt**, Gordon Kindlmann. EuroVis Posters. June 2019. Poster.

2018 [Linting for Visualization: Towards a Practical Automated Visualization Guidance System](#)
Andrew McNutt, Gordon Kindlmann. IEEE VIS VisGuides Workshop. Oct. 2018. Workshop Paper.

Book Chapters

2018 [Data Mining and Computational Modeling of High-Throughput Screening Datasets](#)
Sean Ekins, Alex M Clark, Krishna Dole, Kellan Gregory, **Andrew McNutt**, Anna Coulon Spektor, Charlie Weatherall, Nadia K Litterman, Barry A Bunin. Reporter Gene Assays. 2018.

Patents

2023 [Surfacing visualization mirages](#)
Michael Correll, **Andrew McNutt**. US20210081874A1. Sept. 2023.

Theses

2023 [Understanding and Enhancing JSON-based DSLs for Visualization](#)
Advised by Ravi Chugh. Ph.D. Thesis. University of Chicago. August 2023.

2019 [Design and Analysis of Table Cartograms: Multipurpose Tabular Area-Encoding Displays](#)
Advised by Gordon Kindlmann. Master of Science Thesis. University of Chicago. December 2019.

2014 [Nonequivalent Lagrangian Mechanics](#)
Advised by Nelia Mann. Bachelor of Arts Thesis. Reed College. June 2014.

Panels

2024 Visualization for Villainy. Data Visualization Society Book Club. December. *Panelists:* **Andrew McNutt**. *Org:* Elsie Lee-Robbins, Daria Koshkina

VIS Conference Futures: Community Opinions on Recent Experiences, Challenges, and Opportunities for Hybrid Event Formats. IEEE VIS Panel. Oct. *Org:* Matthew Brehmer, Narges Mahyar. *Panelists:* Tim Dwyer, John Guerra-Gomez, Petra Isenberg, Takayuki Itoh, Elsie Lee-Robbins, **Andrew McNutt**.

VisInPractice: How do we close the gap between data visualization research and practitioners. IEEE VIS Panel. Oct. *Org:* Ayan Biswas, Arjun Srinivasan *Panelists:* Alberto Cairo, Cindy Xiong, **Andrew McNutt**.

Tutorials

2026 Running Online User Studies with the reVISit Framework. EuroVIS. June. Zach Cutler, Jack Wilburn, Hilson Shrestha, Yiren Ding, **Andrew McNutt**, Lane Harrison, Alexander Lex

2025 Running Online User Studies with the reVISit Framework. IEEE VIS. Oct. Jack Wilburn, Hilson Shrestha, Zach Cutler, Yiren Ding, Tingying He, **Andrew McNutt**, Lane Harrison, Alexander Lex

Running Online User Studies with the reVISit Framework. EuroVIS. June. Jack Wilburn, Hilson Shrestha, Zach Cutler, Yiren Ding, Tingying He, **Andrew McNutt**, Lane Harrison, Alexander Lex

Advising / Mentoring

PhD

2024-	Shiyi He	University of Utah
2025-	Zach Cutler (Co-advised with Alexander Lex)	University of Utah
2025-	Nathania (Tia) Josephine	University of Utah

MS

2025-2026	Hima Mynampaty	University of Utah
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Undergraduates

2026	Gabriela Rubio Florian	University of Utah
2024	Ann Baturytski	University of Washington
2024	Shaoming (Ramses) Lee	University of Washington

Teaching

New Course*, Significant Updates[†]

Instructor

2026 Spring	Critical HCI and Visualization	CS6967	University of Utah (10)
Spring	Visualization Seminar	CS7942	University of Utah (9)
2025 Fall	Programming for All 1 [†]	COMP1010	University of Utah (108)
Fall	Visualization Seminar	CS7942	University of Utah (7)
Spring	Critical HCI and Visualization *	CS6967	University of Utah (7)
2024 Fall	Programming for All 1	COMP1010	University of Utah (105)
2021 Winter	Data Visualization For Public Policy	CAPP 30239	UChicago (29 students)
2020 Spring	Data Visualization	CMSC 23900	UChicago (55 students)
Winter	Data Visualization For Public Policy [†]	CAPP 30239	UChicago (24 students)
2016-2017	Ubersersity		Uber
2015-2017	Visualization Eng-ucation		Uber

Teaching Assistant

2023 Spring	Data Visualization and Communication	DATA 22700	UChicago
2022 Winter	Creative Coding	CMSC 11111	UChicago
2021 Summer	Introduction to Creative Coding	CMSC 19111	Pre-College Immersion (HS)
Summer	Art with Science: Creative Coding		Collegiate Scholars Program (HS)
Spring	Creative Coding	CMSC 11111	UChicago
2019 Spring	Data Visualization	CMSC 23900	UChicago
Winter	Data Visualization For Public Policy	CAPP 30239	UChicago
2018 Fall	Computer Science with Applications 1	CAPP 30121	UChicago
Spring	Data Visualization	CMSC 23900	UChicago
Winter	Introduction to Computer Science 1	CMSC 15100	UChicago
2017 Fall	Computer Science with Applications 1	CMSC 12100	UChicago
2012 Fall	General Physics I	Physics 101	Reed College

Service

Program Committee

ACM SIGCHI	Full Papers (26 ★ ²)
EuroVis	Full Papers (25, 26), Short Papers (25, 26)
IEEE VIS	Full Papers (24, 25, 26), VDS (24), VisGuides (22), EduVIS (25), alt.vis (21, 22, 23, 25)
Info+	Full Submissions (25)
LIVE	Main Track (26)

Organizing Committee

IEEE VIS	Web Chair		2026
VISIONs	Organizer	IEEE VIS workshop	2024
Info+	Zine Chair		2023
alt.vis	Lead Organizer	IEEE VIS workshop	2022, 23, 25

Awards Committee

EuroVis	Best Short Paper (25)
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External Reviewing

CHI	Full Papers (21 ★ , 22 ★ ² , 23 ★ ³ , 24 ★, 25), alt.chi (20, 25), Late-Breaking Work (22)
EuroVis	Full Papers (23)
UIST	Full Papers (21 ★, 26)
VIS	Full Papers (20, 21, 22, 23 ★), Short Papers (20, 21), alt.vis (21)
Journal	CG&A (23, 24), JOVI (23-25), TVCG (22, 24, 25), Computer (26)

★ - Special Recognition for Outstanding Review

Professional Service

2025	4 Corners VIS (Regional Professional Meeting)	Organizer
2025	CG&A Special Issue on Critical Vis x2	Guest Editor
2023-	Open Grad Positions in Visualization	Maintainer
2024-	Journal of Visualization and Interaction	Associate Editor
2024	IEEE VIS Satellite	Organizer
2021	EuroVis	Student Volunteer
2019	Open Access VIS / EuroVis	Contributor / Organizer
Session chairing: EuroVis 2025		

Internal Service

2025-	HCC Track	Director
2024-	Curriculum committee	Member
2024-	Admissions committee	Member
2024	Grad Open House (Nov. 15th)	Speaker
2024	Graduate Student Mentoring (Sept 16)	Panelist
2024	Orientation Week	Speaker

Grant Proposal Panels

NSF	2026 (CISE Future Core Panelist), 2026 (DRMS External Expert)
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Doctoral Thesis Committees

2025	Connor Scully-Allison (Adv. Katherine Isaacs)	University of Utah
2026*	Md Dilshadur Rahman (Adv. Paul Rosen)	University of Utah
2026*	Rifat Ara Proma (Adv. Paul Rosen)	University of Utah
2026*	Aashish Panta (Adv. Valerio Pascucci)	University of Utah
2026*	Xinyuan Yan (Adv. Bei Wang)	University of Utah
2027*	Shadmaan Hye (Adv. Katherine Isaacs)	University of Utah
2027*	Sharmin Ahmed (Adv. Sameer Patil)	University of Utah
2028*	Chang Han (Adv. Katherine Isaacs)	University of Utah

*Expected

Professional Development

2025	Vis Summer Camp	VGTC
2023	College Teaching Certificate	UChicago Center for Teaching

Invited Talks

2026	Safer Data Interfaces with Linters University of British Columbia, March, Vancouver, Canada
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Linters as Sociotechnical Systems

AI + Data Science Seminar, University of Utah, January

- 2025 **Linters as Sociotechnical Systems**
King's College London, June, London, England
giCentre, June, London, England
University of Bristol, June, Bristol, England
- Two talks about JSON DSLs for visualization**
University of Utah, Salt Lake City, Utah, March
- 2024 **Safer Data Interfaces with Linters**
Tableau Research, June, 2024, Seattle, Washington.
- 2023 **JSON-based DSLs and the Future of End User Programming**
Microsoft Research, November, 2023, Cambridge, England.
- Projectional Editors for JSON-based DSLs**
Vega Conf, September, 2023, Seattle, Washington.
- Safer Interfaces for Data Visualization**
University of California Berkeley, May, Berkeley, California.
Microsoft Research, April, Cambridge, England (Virtual).
Roux Institute, March, Portland, Maine.
University of Utah, February, Salt Lake City, Utah.
- 2022 **Understanding & Enhancing JSON-based DSL Interfaces for Visualization.**
Shandong University, November 15, Qingdao, China. (Virtual)
IEEE VIS Doctoral Colloquium, October 15, Oklahoma City, Oklahoma.
- How do we know what a visualization is good for? Algebraic Approaches.**
RAMPVIS, August 5, Oxford, England. (Virtual)
- What are Table Cartograms Good for Anyway? An Algebraic Analysis.**
City University London, May 18, London, England. (Virtual)
- Integrated Visualization Editing via Parameterized Declarative Templates.**
University of California: Berkeley, November, Berkeley, California (Virtual)
CHicago Symposium, May, Chicago, Illinois (Virtual)
- 2018 **Design Patterns For Data Visualization in React.**
React Chicago, August 29, 2018, Chicago, Illinois.